

where?

what?

how bad?

fly again?

Stage 0

Screening

$$|x_i - \mu_i|/\sigma_i$$

Mahalanobis

AUROC 0.99



Stage 1

Classification

defect class

GNN

F1 0.69



Stage 2

Characterisation

$$\theta = (c_x, c_y, \ell, s)$$

FMPE posterior

cov. 0.94



Stage 3

Prognosis

phase-field+fatigue

P(survive n)

$\Rightarrow$ OK / REPAIR / RETIRE